**Effat University – College of Engineering**

Department of Computer Science

**CS4074: Big Data Analytics**

# Lab 3 – Supplementary Sheet

MapReduce Programming: Principles Revision & Practice Exercises

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## Part A: MapReduce Coding Principles – A Detailed Revision

This section provides a compact but thorough review of how to think about and write MapReduce programs using Hadoop Streaming with Python.

### A.1 The Golden Rule: Think in Key-Value Pairs

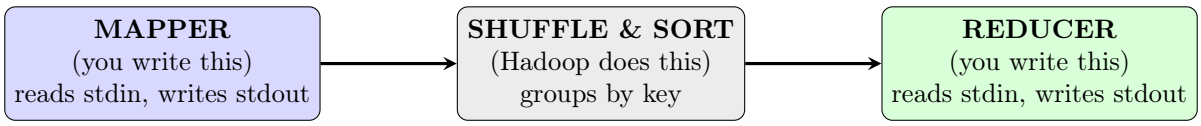
Every MapReduce program revolves around one idea: **data flows as key\tvalue pairs**. Your job as a programmer is to answer two questions:

1. **What is my key?** (The thing I want to group by)
2. **What is my value?** (The thing I want to aggregate)

**Examples of choosing keys and values:**

| Analysis Goal | Key | Value |
|---|---|---|
| Count occurrences of each word | word | 1 |
| Total sales per store | store_name | sale_amount |
| Average grade per course | course_id | grade |
| Max temperature per city | city | temperature |
| List of products per category | category | product_name |

### A.2 The Three Phases – What Happens Where




- **You write:** The Mapper (transforms each input line into key\tvalue) and the Reducer (aggregates all values for each key).
- **Hadoop does:** Reading input splits, distributing to mappers, sorting all mapper output by key, grouping values by key, sending grouped data to reducers, writing final

output to HDFS.

- **You do NOT write:** Any networking, file distribution, sorting, or grouping code.

### A.3 The Mapper – Anatomy & Rules

The mapper is a Python script that reads from `sys.stdin` (one line at a time) and writes to `sys.stdout`.

Mapper Template

```
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    line = line.strip()          # 1. Clean the line

    # 2. Skip invalid lines (headers, empty lines, etc.)
    if not line or line.startswith("header"):
        continue

    # 3. Parse the line (split CSV, extract fields, etc.)
    parts = line.split(",")
    key = parts[INDEX_A]         # Choose your key
    value = parts[INDEX_B]       # Choose your value

    # 4. Emit key-value pair (MUST be separated by TAB)
    print(f"{key}\t{value}")
```

1. **Always use TAB** (`\t`) to separate key and value – not spaces, not commas.
2. **One key-value pair per line** – each `print()` is one output record.
3. **A mapper can emit 0, 1, or many pairs per input line** – e.g., word count emits one pair per word (many per line).
4. **Skip bad data gracefully** – use `try/except` so one bad line doesn't crash the whole job.
5. **No accumulation in the mapper** – the mapper processes each line independently. It has no memory of previous lines.

### A.4 The Reducer – Anatomy & Rules

The reducer receives **sorted** key-value pairs from Hadoop. All pairs with the same key arrive **consecutively** (because Hadoop sorted them). The reducer must detect when the key changes.

Reducer Template - The "Current Key" Pattern

```
#!/usr/bin/env python3
import sys

current_key = None
accumulator = 0           # Could be: sum, count, min, max, list,
    etc.
```

```python
for line in sys.stdin:
    line = line.strip()

    try:
        key, value = line.split('\t', 1)
        value = int(value)      # or float, or keep as string
    except ValueError:
        continue                # Skip malformed lines

    if key == current_key:
        # SAME key: keep accumulating
        accumulator += value    # or: max(), min(), append(), etc.
    else:
        # NEW key detected: output the previous key's result
        if current_key is not None:
            print(f"{current_key}\t{accumulator}")
        # Reset for the new key
        current_key = key
        accumulator = value

# CRITICAL: Don't forget the very last key!
if current_key is not None:
    print(f"{current_key}\t{accumulator}")
```

Forgetting the final `if current_key is not None: print(...)` at the end. Without it, the **last key in the entire dataset** is silently lost. This is the most common MapReduce bug.

### A.5 Adapting the Reducer for Different Aggregations

The reducer template above uses summation. Here's how to adapt it for other operations:

| Aggregation | Initialize | Accumulate |
|---|---|---|
| Sum | `acc = value` | `acc += value` |
| Count | `acc = 1` | `acc += 1` |
| Maximum | `acc = value` | `acc = max(acc, value)` |
| Minimum | `acc = value` | `acc = min(acc, value)` |
| Average | `total=value; count=1`<br>(output: `total/count`) | `total+=value; count+=1` |
| Collect list | `acc = [value]`<br>(output: `",".join(acc)`) | `acc.append(value)` |

### A.6 Local Testing – The Unix Pipeline Trick

**Always test locally before submitting to Hadoop.** The Unix pipe `|` simulates the MapReduce pipeline perfectly:

Local Testing = Simulated MapReduce

```
cat input.txt | python3 mapper.py | sort | python3 reducer.py
#   ^               ^                   ^             ^
#   |               |                   |             |
# HDFS read     MAP phase      SHUFFLE & SORT    REDUCE phase
```

The `sort` command between mapper and reducer is **critical**. Without it, the reducer's "current key" pattern breaks because identical keys won't be consecutive. Hadoop does this sorting automatically, but when testing locally, **you** must add `sort`.

### A.7 Quick Decision Flowchart

When you face a new MapReduce problem, follow this process:

1. **Identify the goal:** What do I want to compute? (total, average, max, list, count...)
2. **Choose the key:** What do I want to group by? (region, city, word, category...)
3. **Choose the value:** What number or data do I need to aggregate?
4. **Write the mapper:** Parse each line → emit `key\tvalue`.
5. **Write the reducer:** Use the "current key" pattern with the right aggregation.
6. **Test locally:** `cat input | python3 mapper.py | sort | python3 reducer.py`
7. **Run on Hadoop:** Use the `hadoop jar $STREAMING_JAR ...` command.

## Part B: Practice Exercises

**Context:** You are a data analyst at a Saudi e-commerce company. You have been given a real transaction log from Q1 2024. Each line records a single order with the customer's city, product category, and order total in SAR (Saudi Riyal).

**Dataset:** Create the following file `ecommerce_sales.csv`:

ecommerce_sales.csv (35 transactions)

```
order_id,date,city,region,category,items,total_sar
1001,2024-01-05,Riyadh,Central,Electronics,2,1450.00
1002,2024-01-05,Jeddah,Western,Fashion,3,320.50
1003,2024-01-06,Dammam,Eastern,Electronics,1,2100.00
1004,2024-01-07,Riyadh,Central,Groceries,5,185.75
1005,2024-01-08,Makkah,Western,Fashion,2,540.00
1006,2024-01-10,Riyadh,Central,Electronics,1,3200.00
1007,2024-01-12,Jeddah,Western,Groceries,8,410.25
1008,2024-01-14,Tabuk,Northern,Electronics,1,890.00
1009,2024-01-15,Abha,Southern,Fashion,4,780.00
1010,2024-01-15,Riyadh,Central,Home,2,1650.00
1011,2024-01-18,Dammam,Eastern,Groceries,3,220.50
1012,2024-01-20,Jeddah,Western,Electronics,2,4500.00
1013,2024-01-22,Riyadh,Central,Fashion,1,125.00
1014,2024-01-25,Khobar,Eastern,Home,3,2800.00
1015,2024-01-28,Madinah,Western,Groceries,6,355.00
1016,2024-02-01,Riyadh,Central,Electronics,1,1800.00
1017,2024-02-03,Jeddah,Western,Fashion,2,690.00
1018,2024-02-05,Dammam,Eastern,Electronics,3,3750.00
1019,2024-02-07,Abha,Southern,Groceries,4,195.50
1020,2024-02-10,Riyadh,Central,Home,1,4200.00
1021,2024-02-12,Makkah,Western,Electronics,2,1350.00
1022,2024-02-14,Tabuk,Northern,Fashion,1,245.00
1023,2024-02-16,Riyadh,Central,Groceries,7,520.00
1024,2024-02-18,Jeddah,Western,Home,2,1980.00
1025,2024-02-20,Dammam,Eastern,Fashion,3,465.00
1026,2024-02-22,Riyadh,Central,Electronics,1,2750.00
1027,2024-02-25,Hail,Northern,Groceries,5,280.00
1028,2024-03-01,Riyadh,Central,Fashion,2,830.00
1029,2024-03-03,Jeddah,Western,Electronics,1,5600.00
1030,2024-03-05,Najran,Southern,Home,2,1200.00
1031,2024-03-08,Riyadh,Central,Groceries,4,310.00
1032,2024-03-10,Dammam,Eastern,Home,1,1550.00
1033,2024-03-12,Jeddah,Western,Groceries,3,175.50
1034,2024-03-15,Riyadh,Central,Electronics,2,3100.00
1035,2024-03-18,Buraidah,Central,Fashion,1,195.00
```

**Task A:** Write a MapReduce program to compute the **total revenue (in SAR) per product category**.
**Task B:** Write a MapReduce program to compute the **average order value per region**.
*Both tasks include full solutions below.*

### Step 1: Identify key and value

- Goal: Total revenue per category

- Key = `category` (field index 4)
- Value = `total_sar` (field index 6)
- Aggregation = **Sum**

category_mapper.py

```python
#!/usr/bin/env python3
"""Mapper: Emit category and sale amount"""
import sys

for line in sys.stdin:
    line = line.strip()
    if not line or line.startswith("order_id"):   # Skip header
        continue
    parts = line.split(",")
    if len(parts) >= 7:
        try:
            category = parts[4]
            total = float(parts[6])
            print(f"{category}\t{total}")
        except (ValueError, IndexError):
            continue
```

category_reducer.py

```python
#!/usr/bin/env python3
"""Reducer: Sum revenue per category"""
import sys

current_key = None
total_revenue = 0.0

for line in sys.stdin:
    line = line.strip()
    try:
        key, value = line.split('\t', 1)
        value = float(value)
    except ValueError:
        continue

    if key == current_key:
        total_revenue += value
    else:
        if current_key is not None:
            print(f"{current_key}\t{total_revenue:.2f}")
        current_key = key
        total_revenue = value

if current_key is not None:
    print(f"{current_key}\t{total_revenue:.2f}")
```

```
cat ecommerce_sales.csv | python3 category_mapper.py | sort | \
    python3 category_reducer.py
```

**Expected Output:**

```
Electronics 30890.00
Fashion 4191.50
Groceries 2652.50
Home  13380.00
```

**Verification:** Electronics = 1450+2100+3200+890+4500+1800+3750+1350+2750+5600+3100 = 30,490... Let's trace carefully:

- Electronics: 1450 + 2100 + 3200 + 890 + 4500 + 1800 + 3750 + 1350 + 2750 + 5600 + 3100 = 30,490.00
- Fashion: 320.50 + 540 + 780 + 125 + 690 + 245 + 465 + 830 + 195 = 4,190.50
- Groceries: 185.75 + 410.25 + 220.50 + 355 + 195.50 + 520 + 280 + 310 + 175.50 = 2,652.50
- Home: 1650 + 2800 + 4200 + 1980 + 1200 + 1550 = 13,380.00

**Step 1: Identify key and value**

- Goal: Average order value per region
- Key = `region` (field index 3)
- Value = `total_sar` (field index 6)
- Aggregation = **Average** (need both sum and count)

region_avg_mapper.py

```python
#!/usr/bin/env python3
"""Mapper: Emit region and order total"""
import sys

for line in sys.stdin:
    line = line.strip()
    if not line or line.startswith("order_id"):
        continue
    parts = line.split(",")
    if len(parts) >= 7:
        try:
            region = parts[3]
            total = float(parts[6])
            print(f"{region}\t{total}")
        except (ValueError, IndexError):
            continue
```

region_avg_reducer.py

```python
#!/usr/bin/env python3
"""Reducer: Compute average order value per region"""
import sys

current_key = None
total_sum = 0.0
count = 0

for line in sys.stdin:
    line = line.strip()
    try:
        key, value = line.split('\t', 1)
        value = float(value)
    except ValueError:
        continue

    if key == current_key:
        total_sum += value
        count += 1
    else:
        if current_key is not None:
            avg = total_sum / count
            print(f"{current_key}\t{avg:.2f}")
        current_key = key
        total_sum = value
        count = 1

if current_key is not None:
    avg = total_sum / count
    print(f"{current_key}\t{avg:.2f}")
```

```
cat ecommerce_sales.csv | python3 region_avg_mapper.py | sort | \
    python3 region_avg_reducer.py
```

**Expected Output:**

```
Central 1522.58
Eastern 1814.33
Northern  471.67
Southern  725.17
Western 1574.25
```

**Reasoning:**

- Central: 13 orders, total = 19,793.50 SAR → avg = 1,522.58
- Eastern: 6 orders, total = 10,885.50 SAR → avg = 1,814.25
- Northern: 3 orders, total = 1,415.00 SAR → avg = 471.67
- Southern: 3 orders, total = 2,175.50 SAR → avg = 725.17
- Western: 10 orders, total = 15,921.25 SAR → avg = 1,592.13

**Average requires two accumulators** (sum and count) in the reducer. This is also why the reducer for average **cannot** be reused as a combiner – a combiner would average partial averages, which is mathematically incorrect. (Recall Reflection 5.1 from Lab 3.)

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**Context:** The General Vehicles Syndicate has collected bus trip records from the 1445 Hajj season. Each record logs a single bus trip between locations in the Makkah region.

**Dataset:** Create the file `hajj_transport.csv`:

hajj_transport.csv

```
trip_id,date,origin,destination,bus_capacity,passengers,duration_min
T001,2024-06-14,Jeddah_Airport,Makkah_Hotel,50,48,95
T002,2024-06-14,Jeddah_Airport,Makkah_Hotel,50,50,110
T003,2024-06-14,Madinah,Makkah_Hotel,50,45,320
T004,2024-06-15,Makkah_Hotel,Mina,50,50,45
T005,2024-06-15,Makkah_Hotel,Mina,50,49,55
T006,2024-06-15,Makkah_Hotel,Mina,50,50,40
T007,2024-06-15,Makkah_Hotel,Mina,50,47,60
T008,2024-06-16,Mina,Arafat,50,50,35
T009,2024-06-16,Mina,Arafat,50,50,30
T010,2024-06-16,Mina,Arafat,50,48,45
T011,2024-06-16,Mina,Arafat,50,50,40
T012,2024-06-16,Arafat,Muzdalifah,50,50,90
T013,2024-06-16,Arafat,Muzdalifah,50,49,85
T014,2024-06-16,Arafat,Muzdalifah,50,50,95
T015,2024-06-17,Muzdalifah,Mina,50,50,25
T016,2024-06-17,Muzdalifah,Mina,50,48,30
T017,2024-06-17,Muzdalifah,Mina,50,50,20
T018,2024-06-18,Mina,Makkah_Hotel,50,46,50
T019,2024-06-18,Mina,Makkah_Hotel,50,50,55
T020,2024-06-19,Makkah_Hotel,Jeddah_Airport,50,44,90
T021,2024-06-19,Makkah_Hotel,Jeddah_Airport,50,50,85
T022,2024-06-19,Makkah_Hotel,Madinah,50,38,340
T023,2024-06-14,Jeddah_Airport,Makkah_Hotel,50,50,100
T024,2024-06-15,Makkah_Hotel,Mina,50,50,50
T025,2024-06-16,Mina,Arafat,50,49,38
```

## Exercise 2 – Tasks (Write Your Own Solution)

**Task A:** Write a MapReduce program to compute the **total number of passengers transported on each route** (route = origin→destination pair).
*Hint:* Your key should combine origin and destination, e.g., `Mina->Arafat`.

**Task B:** Write a MapReduce program to find the **longest trip duration (in minutes) for each date**.

**a) Task A – Mapper code: b) Task A – Reducer code: c) Task A – Local test**

Task A

```
#!/usr/bin/env python3
"""Mapper: Emit route and passengers"""
import sys

for line in sys.stdin:
    line = line.strip()
    if not line or line.startswith("trip_id"):
        continue
    parts = line.split(",")
    if len(parts) >= 7:
        try:
            origin = parts[2].strip()
            destination = parts[3].strip()
            passengers = int(parts[5])
            route = f"{origin}->{destination}"
            print(f"{route}\t{passengers}")
        except (ValueError, IndexError):
            continue
```

```
#!/usr/bin/env python3
"""Reducer: Sum passengers per route"""
import sys

current_route = None
total_passengers = 0

for line in sys.stdin:
    line = line.strip()
    try:
        route, value = line.split("\t", 1)
        value = int(value)
    except ValueError:
        continue

    if route == current_route:
        total_passengers += value
    else:
        if current_route is not None:
            print(f"{current_route}\t{total_passengers}")

current_route = route
        total_passengers = value

if current_route is not None:

print(f"{current_route}\t{total_passengers}")
```

```
chmod +x
passengers_mapper.py
chmod +x
passengers_reducer.py
```

**command and output: d) Task B – Mapper code: e) Task B – Reducer code: f)**

```
cat hajj_transport.csv | python3 passengers_mapper.py | sort | python3 passengers_reducer.py
```

Task B

Mapper code

```python
#!/usr/bin/env python3
"""Mapper: Emit date and duration"""
import sys

for line in sys.stdin:
    line = line.strip()
    if not line or line.startswith("trip_id"):
        continue
    parts = line.split(",")
    if len(parts) >= 7:
        try:
            date = parts[1].strip()
            duration = int(parts[6])
            print(f"{date}\t{duration}")
        except (ValueError, IndexError):
            continue
```

```
Arafat->Muzdalifah   149
Jeddah_Airport->Makkah_Hotel   148
Makkah_Hotel->Jeddah_Airport   94
Makkah_Hotel->Mina   249
Mina->Arafat  147
Muzdalifah->Mina  148
Madinah->Makkah_Hotel  45
```

Reducer code

```python
#!/usr/bin/env python3
"""Reducer: Find max duration per date"""
import sys

current_date = None
max_duration = 0

for line in sys.stdin:
    line = line.strip()
    try:
        date, value = line.split("\t", 1)
        value = int(value)
    except ValueError:
        continue

    if date == current_date:
        if value > max_duration:
            max_duration = value
    else:
        if current_date is not None:
            print(f"{current_date}\t{max_duration}")
        current_date = date
        max_duration = value

if current_date is not None:
    print(f"{current_date}\t{max_duration}")
```

**Task B – Local test command and output:**

```
chmod +x duration_mapper.py
chmod +x duration_reducer.py

cat hajj_transport.csv | python3 duration_mapper.py | sort | python3 duration_reducer.py
```

```
2024-06-14   320
2024-06-15   60
2024-06-16   95
2024-06-17   30
2024-06-18   55
2024-06-19   340
```

**Context:** The Effat University registrar has exported enrollment data for the Fall 2024 semester. Each record represents one student's enrollment in one course.

**Dataset:** Create the file `enrollments.csv`:

enrollments.csv

```
student_id,student_name,department,course_code,course_name,credits,grade
S101,Noura,CS,CS4074,Big Data Analytics,3,A
S102,Sara,CS,CS4074,Big Data Analytics,3,B+
S103,Lama,CS,CS4074,Big Data Analytics,3,A
S104,Reem,ECE,CS4074,Big Data Analytics,3,B
S105,Dana,CS,CS4082,Machine Learning,3,A+
S106,Noura,CS,CS4082,Machine Learning,3,A
S107,Huda,CS,CS4082,Machine Learning,3,B+
S108,Amal,ECE,ECE333,Intro to AI,3,B
S109,Fatima,ECE,ECE333,Intro to AI,3,A
S110,Rawan,ECE,ECE333,Intro to AI,3,C+
S111,Maha,ECE,ECE333,Intro to AI,3,B+
S112,Sara,CS,CS4083,NLP,3,A
S113,Lama,CS,CS4083,NLP,3,A+
S114,Huda,CS,CS4083,NLP,3,B
S115,Dana,CS,CS4086,Advanced AI,3,A
S116,Noura,CS,CS4086,Advanced AI,3,A+
S117,Reem,ECE,CS4086,Advanced AI,3,B+
S118,Amal,ECE,ECE301,Signal Processing,3,C
S119,Fatima,ECE,ECE301,Signal Processing,3,B
S120,Maha,ECE,ECE301,Signal Processing,3,B+
S121,Rawan,ECE,ECE301,Signal Processing,3,A
S122,Sara,CS,CS4074,Big Data Analytics,3,A+
S123,Huda,CS,CS4074,Big Data Analytics,3,B
S124,Dana,CS,CS4083,NLP,3,A
S125,Fatima,ECE,ECE333,Intro to AI,3,A+
```

## Exercise 3 – Tasks (Write Your Own Solution)

**Task A:** Write a MapReduce program to count the **number of students enrolled in each course** (by course_code).

**Task B:** Write a MapReduce program to compute the **average GPA per department**. Use the following grade-to-GPA conversion: A+=4.0, A=4.0, B+=3.5, B=3.0, C+=2.5, C=2.0, D=1.0, F=0.0.

*Hint for Task B:* The mapper should convert the letter grade to a numeric GPA before emitting it.

**Task C (Bonus):** Write a MapReduce program to find the **student with the highest GPA in each department**. If there is a tie, list all tied students.

**a) Task A – Mapper code: b) Task A – Reducer code: c) Task A – Local test**

Task A

```python
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    line = line.strip()
    if not line or line.startswith("student_id"):
        continue
    parts = line.split(",")
    if len(parts) >= 6:
        try:
            course_code = parts[3].strip()
            print(f"{course_code}\t1")
        except IndexError:
            continue
```

```python
import sys
current_course = None
count = 0

for line in sys.stdin:
    line = line.strip()
    try:
        course, value = line.split("\t", 1)
        value = int(value)
    except ValueError:
        continue

    if course == current_course:
        count += value
    else:
        if current_course is not None:
            print(f"{current_course}\t{count}")
        current_course = course
        count = value

if current_course is not None:
    print(f"{current_course}\t{count}")
```

```
cat enrollments.csv | python3 course_count_mapper.py |
sort | python3 course_count_reducer.py
```

Task B

**command and output: d) Task B – Mapper code: e) Task B – Reducer code:**

```
#!/usr/bin/env python3
import sys

# Grade to GPA mapping
grade_to_gpa = {
    "A+": 4.0, "A": 4.0,
    "B+": 3.5, "B": 3.0,
    "C+": 2.5, "C": 2.0,
    "D": 1.0, "F": 0.0
}
for line in sys.stdin:
    line = line.strip()
    if not line or line.startswith("student_id"):
        continue
    parts = line.split(",")
    if len(parts) >= 7:
        try:
            department = parts[2].strip()
            grade = parts[6].strip()
            gpa = grade_to_gpa.get(grade, 0.0)
            print(f"{department}\t{gpa}")
        except IndexError:
            continue
```

```
#!/usr/bin/env python3
import sys

current_dept = None
total_gpa = 0.0
count = 0

for line in sys.stdin:
    line = line.strip()
    try:
        dept, value = line.split("\t", 1)
        value = float(value)
    except ValueError:
        continue

    if dept == current_dept:
        total_gpa += value
        count += 1
    else:
        if current_dept is not None:
            avg_gpa = total_gpa / count
            print(f"{current_dept}\t{avg_gpa:.2f}")
        current_dept = dept
        total_gpa = value
        count = 1
if current_dept is not None:
    avg_gpa = total_gpa / count
    print(f"{current_dept}\t{avg_gpa:.2f}")
```

**f) Task B – Local test command and output: g) Task C (Bonus) – Mapper and**

```
cat enrollments.csv | python3 gpa_mapper.py | sort | python3 gpa_reducer.py
```

```
CS    3.86
ECE   3.06
```

Task C

**Reducer code: h) Task C (Bonus) – Output:**

```
#!/usr/bin/env python3
"""Mapper: Emit department, student_name, GPA"""
import sys

grade_to_gpa = {
    "A+": 4.0, "A": 4.0,
    "B+": 3.5, "B": 3.0,
    "C+": 2.5, "C": 2.0,
    "D": 1.0, "F": 0.0
}

for line in sys.stdin:
    line = line.strip()
    if not line or line.startswith("student_id"):
        continue
    parts = line.split(",")
    if len(parts) >= 7:
        try:
            dept = parts[2].strip()
            student = parts[1].strip()
            grade = parts[6].strip()
            gpa = grade_to_gpa.get(grade, 0.0)
            print(f"{dept}\t{student}\t{gpa}")
        except IndexError:
            continue
```

```
#!/usr/bin/env python3
"""Reducer: Find student(s) with highest GPA per
department"""
import sys

current_dept = None
max_gpa = -1
top_students = []

for line in sys.stdin:
    line = line.strip()
    try:
        dept, student, gpa = line.split("\t")
        gpa = float(gpa)
    except ValueError:
        continue

    if dept == current_dept:
        if gpa > max_gpa:
            max_gpa = gpa
            top_students = [student]
        elif gpa == max_gpa:
            top_students.append(student)
    else:
        if current_dept is not None:
            print(f"{current_dept}\t{',
'.join(top_students)}\t{max_gpa:.2f}")
        current_dept = dept
        max_gpa = gpa
        top_students = [student]

if current_dept is not None:
    print(f"{current_dept}\t{',
'.join(top_students)}\t{max_gpa:.2f}")
```

---

```
cat enrollments.csv | python3 top_student_mapper.py | sort | python3 top_student_reducer.py
```

```
CS    Dana,Noura,Lama    4.00
ECE   Rawan,Fatima       4.00
```

**Context:** In the real world, data doesn't come pre-formatted in a lab handout. Data engineers must find appropriate datasets, understand their structure, clean them, and process them at scale. In this exercise, you will experience the full pipeline: **find** → **download** → **explore** → **design** → **implement** → **analyze** → **report.**

### Step 1: Choose and Download a Dataset

Select **one** dataset from the sources below (or find your own). The dataset must be:

- In CSV or text format (or convertible to CSV)
- At least **1,000 rows**
- Containing at least **one categorical column** (to group by) and at least **one numerical column** (to aggregate)

| Source | URL | Suggested Datasets |
|---|---|---|
| Kaggle | https://www.kaggle.com/datasets | Global Superstore Sales, World University Rankings, FIFA Players |
| Saudi Open Data | https://data.gov.sa | Health facilities, Education statistics, Traffic data |
| UCI ML Repository | https://archive.ics.uci.edu | Adult Census Income, Online Retail, Air Quality |
| data.gov | https://data.gov | US Baby Names, Airport Delays, Food Inspections |
| World Bank | https://data.worldbank.org | GDP by country, Population growth, Education indicators |

If you're unsure what to choose, download the **Global Superstore Sales** dataset from Kaggle (∼10,000 rows). It has clear categorical columns (Country, Category, Segment) and numerical columns (Sales, Profit, Quantity) — perfect for MapReduce.

### Step 2: Upload to HDFS

Download and Upload

```
# Download your dataset (example: from Kaggle after login)
# Move the downloaded file to your home directory
mv ~/Downloads/your_dataset.csv ~/dataset.csv

# Inspect the file
head -5 ~/dataset.csv          # See the first 5 lines
wc -l ~/dataset.csv            # Count total rows
du -h ~/dataset.csv             # Check file size

# Upload to HDFS
hdfs dfs -mkdir -p /user/student/lab3/exercise4_input
hdfs dfs -put ~/dataset.csv /user/student/lab3/exercise4_input/
```

### Step 3: Design and Implement Two MapReduce Analyses

You must design and implement **two different MapReduce programs** on your chosen dataset. Each program must use a different aggregation type. Choose from:

| Aggregation Type | Example Question |
|---|---|
| Sum | What is the total sales per country? |
| Count | How many records per category? |
| Average | What is the average score per department? |
| Max / Min | What is the highest temperature per city? |
| Collect / List | Which products belong to each category? |

Your two analyses must use **different keys** and **different aggregation types**. For example, if Analysis 1 computes "total sales per country" (key=country, aggregation=sum), then Analysis 2 could be "average profit per category" (key=category, aggregation=average).

### Step 4: Write a Processing Report

After implementing and running your MapReduce programs, write a **1–2 page report** that covers the following sections:

**Exercise 4 – Report Structure**

Your report must include the following sections:

1. **Dataset Description** (5 lines minimum)
   - Name of the dataset and source URL
   - Number of rows and columns
   - Description of each column (name, type, meaning)
   - Any data quality issues observed (missing values, inconsistent formats, etc.)

2. **Analysis Design** (for each of the 2 analyses)
   - The question you are answering
   - What you chose as the key and why
   - What you chose as the value and why
   - What aggregation type you used

3. **Implementation**
   - Full mapper and reducer code for both analyses
   - Any data cleaning steps performed in the mapper (e.g., skipping headers, handling missing values, type conversions)
   - The local test command and its output
   - The Hadoop execution command

**4. Results & Discussion**

- Output of both MapReduce programs (paste the actual results)
- A brief interpretation: What did you learn from the data? Were there any surprising findings?
- Manually verify at least 2 output rows by tracing through the original data

**5. Challenges & Lessons Learned** (5 lines minimum)

- What difficulties did you encounter? (e.g., encoding issues, delimiter problems, dirty data)
- How did you solve them?
- What would you do differently next time?

## Exercise 4 – Deliverables Checklist

Submit the following items:

| # | Item | Included? |
|---|---|---|
| 1 | Dataset file (or link if >10 MB) | ☐ |
| 2 | Screenshot: `head -5` and `wc -l` of your dataset | ☐ |
| 3 | Analysis 1: mapper.py + reducer.py code | ☐ |
| 4 | Analysis 1: local test output screenshot | ☐ |
| 5 | Analysis 1: Hadoop output (first 15 lines) | ☐ |
| 6 | Analysis 2: mapper.py + reducer.py code | ☐ |
| 7 | Analysis 2: local test output screenshot | ☐ |
| 8 | Analysis 2: Hadoop output (first 15 lines) | ☐ |
| 9 | Screenshot: YARN Web UI showing completed jobs | ☐ |
| 10 | Written report (1–2 pages covering all 5 sections above) | ☐ |

In industry, no one gives you a clean 15-row CSV. Real datasets are messy, large, and undocumented. The skills you practice here — exploring unfamiliar data, handling edge cases in your mapper, verifying results manually, and communicating findings clearly — are exactly what employers look for in data engineers and data scientists. This exercise bridges the gap between **lab exercises** and **real-world data processing**.

**End of Supplementary Sheet**